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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/560,159	11/10/2023	Norman Haag	BOSC.P14382US/1001273842	3216

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EXAMINER
YICK, JORDAN WAN

ART UNIT	PAPER NUMBER
2612	

NOTIFICATION DATE	DELIVERY MODE
07/30/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/560,159	Applicant(s) Haag et al.	
	Examiner JORDAN W YICK	Art Unit 2612	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10 November 2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 11-20 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 11-14 and 16-20 is/are rejected.

8) ☑ Claim(s) 15 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10 November 2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 11/10/2023, 12/30/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Claim Rejections - 35 USC § 103

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 11,13-14, 18-20 are rejected under 35 U.S.C. 103 as being unpatentable over Parchami (US 20200232799 A1), hereinafter Parchami, in view of Kaganovich (US 20140118716 A1), hereinafter Kaganovich, and Helsloot (US 20210033711 A1), hereinafter Helsloot.

Regarding claim 11, Parchami teaches a method for determining concealed objects in a 3D point cloud representing an environment, the method comprising the follow steps:

producing a 3D point cloud using an active sensor, wherein each point of the 3D point cloud

represents a distance measurement by the active sensor (Fig. 4A, paragraph 53, wherein a 3D representation of a volume obtained from a LIDAR sensor including location coordinates such as longitudinal, lateral, and elevation coordinates is interpreted as representing a distance measurement by the active sensor); determining **shadows within the background region of the 3D point cloud, wherein the shadows are regions within the background region at which there are substantially no points of the 3D point cloud** (Fig. 6, paragraph 71-74, wherein determining dimensions of low-density areas which may be shadows casted by an object is interpreted as determining shadows within a region of a 3D point cloud); and determining**, in an environment of the active sensor, an object in the active sensor**, by identifying, in the background region of the 3D point cloud, **at least one shadow which cannot be attributed to an object in a foreground of the 3D point cloud** (Fig. 6, paragraph 74-75, wherein an invisible object may be detected based on a low-density area, in response to not detecting a second object that would cast that shadow, which is interpreted as determining an object based on a shadow that cannot be attributed to an object in the foreground of the 3D point cloud).

Parchami does not teach determining **a background region within the 3D point cloud**; and determining **an object which is concealed by crosstalk.**

Kaganovich teaches determining **a background region within the 3D point cloud** (Fig. 3B, segmenting a target object from the background for a LIDAR range image interpreted as determining a background region within a 3D point cloud).

Helsloot teaches determining **an object which is concealed by crosstalk in the active sensor** (Fig. 3A-3B, paragraph 67-68, an object in a point cloud that may be obscured by an artefact formed from vertical and horizontal crosstalk).

It would be obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to have modified Parchami to incorporate the teachings of Kaganovich and Helsloot for this method for determining concealed objects in a 3D point cloud. Parchami discusses detecting low density areas in a point cloud, and determining if that corresponds to a second detected object, or if it represents an invisible or concealed physical object, for the purposes of implementing a navigation system for a vehicle. Kaganovich discusses segmenting a LIDAR frame into a target object and a background, for the purposes of better tracking an object while it's moving in areas that may be partially obscured. Helsloot discusses methods for reducing optical crosstalking within a LIDAR system as a result of heavy reflections or glare, in order to better detect objects. As the references discuss analogous art, and discuss non conflicting ways to improve object detection for a point cloud generated by a LIDAR system, it would be obvious to combine them.

Regarding claim 13, Parchami in view of Kaganovich and Helsloot disclose the method of claim 11. Additionally, Parchami teaches the method according to claim 11, wherein **a type** and/or **a size** and/or **a position** of the concealed object is determined **based on a characteristic of the shadow produced by the object** (Paragraph 84-85, wherein a low-density area is interpreted as a shadow, and identifying the location and/or dimensions of a physical object based on an associated low-density area is interpreted as determining the type, size, and/or position of a concealed object based on the characteristics of its shadow).

Regarding claim 14, Parchami in view of Kaganovich and Helsloot disclose the method of claim 13. Additionally, Parchami teaches the method according to claim 13, wherein **the information about the shadow,** determined in the background region**, of the object concealed in the foreground is used to add points representing the concealed object to the 3D point cloud in a region of the foreground in which the concealed object was determined** (Fig. 9A-9B, paragraph 75-76, wherein an unexpected low-density area may result from an invisible object, and wherein the processor broadcasts object data, which is defined as data taken from a point cloud; paragraph 84-85, wherein the processor updates the object data based on identifying a physical object in an unexpected low-density area, which suggests adding points to the point cloud object data representing a concealed object in the region the object was determined).

Regarding claim 18, Parchami in view of Kaganovich and Helsloot disclose the method of claim 14. Additionally, Parchami teaches the method according to claim 14, wherein information about the concealed object and/or the 3D point cloud to which the concealed object is **added is taken into account during environment recognition in an environment recognition system** (Paragraph 76, wherein broadcasted object data which includes object data obtained from a LIDAR and concealed object data is broadcasted to a computer, vehicle, or processor; paragraph 42-44 wherein a processor receives map data including data collected from a LIDAR or other sensors to describe the location, dimensions, and objects in a mapped area, which is interpreted as an environment recognition system).

Regarding claim 19, Parchami in view of Kaganovich and Helsloot disclose the method of claim 11. Additionally, Parchami teaches the method according to claim 11, wherein the active sensor is **a lidar sensor of an environment sensing system of a transportation arrangement** (Fig. 3, paragraph 37, LIDAR sensor that may be used to operate a vehicle, which is interpreted as being a part of a transportation arrangement).

Regarding claim 20, Parchami teaches A device for determining concealed objects in a 3D point cloud representing an environment, comprising: **an active sensor** (Fig 1-2, paragraph 36-37, including one or more sensors); and **an evaluation unit** (Fig. 2, paragraph 40, processor programmed to determine a type of objects interpreted as an evaluation unit); wherein the evaluation unit is configured to determine concealed objects in a 3D point cloud representing an environment, the evaluation unit configured to: **produce a 3D point cloud using an active sensor, wherein each point of the 3D point cloud represents a distance measurement by the active sensor** (Fig. 4A, paragraph 53, wherein a 3D representation of a volume obtained from a LIDAR sensor including location coordinates such as longitudinal, lateral, and elevation coordinates is interpreted as representing a distance measurement by the active sensor); determine **shadows within the background region of the 3D point cloud, wherein the shadows are regions within the background region at which there are substantially no points of the 3D point cloud** (Fig. 6, paragraph 71-74, wherein determining dimensions of low-density areas which may be shadows casted by an object is interpreted as determining shadows within a region of a 3D point cloud); and determine**, in an environment of the active sensor, an object in the active sensor**, by identifying, in the background region of the 3D point cloud, **at least one**

shadow which cannot be attributed to an object in a foreground of the 3D point cloud (Fig. 6, paragraph 74-75, wherein an invisible object may be detected based on a low-density area, in response to not detecting a second object that would cast that shadow, which is interpreted as determining an object based on a shadow that cannot be attributed to an object in the foreground of the 3D point cloud).

Parchami does not teach determining **a background region within the 3D point cloud**; and determining **an object which is concealed by crosstalk.**

Kaganovich teaches determining **a background region within the 3D point cloud** (Fig. 3B, segmenting a target object from the background for a LIDAR range image interpreted as determining a background region within a 3D point cloud).

Helsloot teaches determining **an object which is concealed by crosstalk in the active sensor** (Fig. 3A-3B, paragraph 67-68, an object in a point cloud that may be obscured by an artefact formed from vertical and horizontal crosstalk).

Claim 12 is rejected under 35 U.S.C. 103 as being unpatentable over Parchami in view of Kaganovich and Helsloot as applied to claim 11 above, and further in view of Zhao (J. Zhao, H. Xu, X. Xia and H. Liu, "Azimuth-Height Background Filtering Method for Roadside LiDAR Data," 2019 IEEE Intelligent Transportation Systems Conference (ITSC), Auckland, New Zealand, 2019, pp. 2421-2426, doi: 10.1109/ITSC.2019.8917369.), hereinafter Liu.

Regarding claim 12, Parchami in view of Kaganovich and Helsloot disclose the method of claim 11. Additionally, Zhao teaches the method according to claim 11, wherein the background

region is identified within the 3D point cloud by: identifying, **for each solid angle of the 3D point cloud, a last echo resulting from a scanning signal emitted by the active sensor** (Section II, wherein processed data frame of a LIDAR is generated from a previously completed scan, which is interpreted as the last echo resulting from a scanning signal; Section III, wherein points from processed data frames are filtered based on their azimuth angle to identify background regions, which suggests the last echo from a scanning signal is used to identify background regions).

Zhao does not teach determining contiguous areas within the 3D point cloud that: have **a predefined minimum distance from the active sensor**, and/or **are oriented within a predefined angle range in relation to a main axis of the active sensor**, and/or have **a predefined minimum size**.

Parchami teaches determining contiguous areas within the 3D point cloud that are **oriented within a predefined angle range in relation to a main axis of the active sensor** (Fig. 1-2, paragraph 35-36, wherein a geographic area defined based on a detection field of view of a fixed sensor is interpreted as an area that can include a background area being oriented within a predefined angle range related to a main axis of the sensor).

Parchami and Zhao do not teach determining contiguous areas within the 3D point cloud that: have **a predefined minimum distance from the active sensor**, and/or have **a predefined minimum size**.

Kaganovich teaches determining contiguous areas within the 3D point cloud that: **have a predefined minimum distance from the active sensor** (paragraph 43, segmentation process for determining a background region may have a distance threshold to determine object

detection), and/or **have a predefined minimum size** (paragraph 29, segmentation process for determining a background region may have a predetermined size threshold to identify target images).

It would be obvious before the effective filing date of the claimed invention to have modified Parchami to incorporate the teachings of Kaganovich, Helsloot, and Zhao for this method of determining background regions of a point cloud. Parchami discusses detecting low density areas in a point cloud, and determining if that corresponds to a second detected object, or if it represents an invisible or concealed physical object, for the purposes of implementing a navigation system for a vehicle. Kaganovich discusses segmenting a LIDAR frame into a target object and a background, for the purposes of better tracking an object while it's moving in areas that may be partially obscured. Similarly, Zhao discusses using the azimuth and angle of points from LIDAR sensors in order to filter out background points, for the purposes of filtering out unrelated information. Helsloot discusses methods for reducing optical crosstalking within a LIDAR system as a result of heavy reflections or glare, in order to better detect objects. As all three references discuss ways to better identify objects from a LIDAR point cloud, and Zhao and Kaganovich both discuss ways to differentiate backgrounds from objects, it would be obvious to combine them.

Claims 16-17 are rejected under 35 U.S.C. 103 as being unpatentable over Parchami in view of Kaganovich and Helsloot as applied to claim 11 above, and further in view of Maheshwari (US 10551485 B1), hereinafter Maheshwari.

Regarding claim 16, Parchami in view of Kaganovich and Helsloot disclose the method of claim 11. Additionally, Maheshwari teaches the method according to claim 11, wherein, using **a further sensor of a different type than the active sensor and/or map material related to a current environment** (Fig. 12, Col 32 lines 4-15, wherein a second set of sensor data captured by a second sensor such as a RGB or stereo camera is interpreted as a different type of sensor and/or map material), **a plausibility check** (Fig. 12, Col. 32 lines 16-59, wherein the first and second sets of sensor data are compared together to identify regions in each set of sensor data that correspond to the same portion of an environment, which is interpreted as a plausibility test) is made of: **the determined background region**, and/or **a presence of the concealed object**, and/or **a type** and/or **a size** and/or **a position of the concealed object**, and/or **existing crosstalk** (Fig. 12, Col. 32 line 60-Col. 33 line 36, wherein the first and second set of sensor data's shapes are compared to verify the "true" shape of the object, wherein the shape of the object is interpreted as including the object's presence, type, size, and/or position, and to determine if the point cloud object is distorted, wherein distortions are interpreted as including distortions caused by crosstalk).

It would be obvious before the effective filing date of the claimed invention to have modified Parchami to incorporate the teachings of Kaganovich, Helsloot, and Maheshwari for this method of performing a plausibility check on sensor data. Parchami discusses detecting low density areas in a point cloud, and determining if that corresponds to a second detected object, or if it represents an invisible or concealed physical object, for the purposes of implementing a navigation system for a vehicle. Kaganovich discusses segmenting a LIDAR frame into a target object and a background, for the purposes of better tracking an object while it's moving in

areas that may be partially obscured. Helsloot discusses methods for reducing optical crosstalking within a LIDAR system as a result of heavy reflections or glare, in order to better detect objects. Likewise, Maheshwari discusses mapping point cloud data to objects in order to better detect those objects while they're moving. Additionally, Parchami, Helsloot, and Maheshwari, each discuss using their invention for better improving autonomous vehicles by better detecting obstacles via LIDAR. Similarly, Parchami, Kaganovich, and Maheshwari each discuss improvements for detecting vehicles using LIDAR, with Kaganovich and Maheshwari focusing on detecting moving vehicles. As all four references discuss processing LIDAR data in order to better detect objects, and especially in relation to using LIDAR for vehicles, it would be obvious to combine them.

Regarding claim 17, Parchami in view of Kaganovich and Helsloot disclose the method of claim 11. Additionally, Maheshwari teaches the method according to claim 11, wherein the determination of the shadows in the background region of the 3D point cloud takes place **using a machine learning method** (Col. 16, lines 44-56, wherein neural works could be utilized to identify any cluster of points that meet certain criteria, which suggests including identifying shadow regions).

The motivation to combine would be the same as that set forth in claim 16.

Allowable Subject Matter

Claim 15 objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to JORDAN W YICK whose telephone number is (571)272-4063. The examiner can normally be reached M-F 8-5.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Said Broome can be reached at (571) 272-2931. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/JORDAN WAN YICK/
Examiner, Art Unit 2612

/Said Broome/
Supervisory Patent Examiner, Art Unit 2612